Scalability & Future Plan

Date	28 October 2026
Team ID	EN25CS3011046
Project Name	Formula 1 Race Intelligence Dashboard

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Performance of external embedded/published views depends on Tableau Public and network response.	Slow loads can interrupt a live demo.	Medium
2	Dashboard calculations and some chart axes/null indicators need validation.	Incorrect interpretation could weaken analytical conclusions.	High
3	Dataset provenance and update cadence are not documented in supplied evidence.	Reproducibility and future refreshes may be difficult.	High
4	No measured concurrent-user, load, or response-time results were supplied.	Capacity claims cannot yet be verified.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Small public-facing Flask site; no verified concurrent-user benchmark supplied.	Measure expected traffic; use suitable hosting capacity and caching for static assets if needed.
2	Data Storage	Analytical data is used through the Tableau workbook; database configuration was not evidenced.	Document data source and schema. If data volume grows, evaluate a managed database/warehouse and scheduled refreshes.
3	Performance	Tableau Public and the hosted web app are separate services; no timings supplied.	Benchmark website load, dashboard render, and filter response; simplify expensive calculations and reduce unnecessary marks if needed.
4	Security	Public project website and public repository are available.	Do not commit secrets; keep dependencies updated, validate user input if forms are added, and review repository contents before publishing.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Validate chart calculations, null values, axis fields, and add a data dictionary/source attribution.	Next 2–4 weeks after review	Improve trust and reproducibility.

Phase 3	Add documented dataset refresh process and expand selected driver/team/season analyses where data supports it.	1–3 months	Keep analysis maintainable and extendable.
Phase 4	Run performance and usability tests; improve accessibility, responsive layout, and error/fallback messaging.	3–6 months	Improve reliability and user experience.

Timelines are proposed planning estimates, not committed delivery dates.